DRAFT — UNOFFICIAL — NOT FOR OPERATIONAL USE

CONCEPTS GUIDE

TM-50E

USAREUR-AF

CONCEPTS GUIDE — TM-50E COMPANION

ADVANCED KNOWLEDGE MANAGER

HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

DRAFT — NOT FOR OFFICIAL USE. FOR TRAINING PLANNING PURPOSES ONLY.

11 MARCH 2026

DRAFT — UNOFFICIAL — NOT FOR OPERATIONAL USE

DRAFT

CONCEPTS GUIDE — TM-50E COMPANION

ADVANCED KNOWLEDGE MANAGER

MAVEN SMART SYSTEM (MSS)

HEADQUARTERS, UNITED STATES ARMY EUROPE AND AFRICA Wiesbaden, Germany 2026

PURPOSE: This guide extends the mental models established in the TM-40E Concepts Guide to advanced knowledge management on MSS. Prerequisite: TM-40E Concepts Guide and TM-40E qualification.

DISTRIBUTION RESTRICTION: Approved for public release; distribution is unlimited.

HOW TO USE THIS GUIDE

This guide is conceptual, not procedural. It does not teach you how to build a Workshop application or configure an Ontology Object Type — TM-40E and TM-50E cover those tasks. This guide teaches you how to think at the enterprise level: how to reason about knowledge systems that span multiple organizations, survive personnel rotation, integrate AI augmentation responsibly, and produce measurable outcomes for the commands they serve.

Read this guide before beginning TM-50E tasks. Return to relevant sections when you encounter design decisions that have no obvious right answer — enterprise KM is full of them. The goal is not to memorize frameworks but to develop the professional judgment that TM-50E demands.

TABLE OF CONTENTS

1. From Knowledge Manager to Organizational Learning Architect
2. Enterprise Knowledge Architecture
3. AI-Assisted Knowledge Management — The MSS Opportunity

DRAFT

4. Knowledge Quality Architecture
5. Cross-Domain Knowledge Integration
6. Coalition Knowledge Management
7. Measuring Knowledge System Effectiveness
8. Knowledge Lifecycle Management at Scale
9. Advanced Failure Modes — What TM-50E KMs Get Wrong

SECTION 1 — FROM KNOWLEDGE MANAGER TO ORGANIZATIONAL LEARNING ARCHITECT

1-1. The Shift in Role

BLUF: At TM-50E level, you are no longer maintaining a knowledge system. You are designing how the organization learns — a fundamentally different mandate with fundamentally different consequences if executed poorly.

A TM-40E KM is a skilled technician and curator. They build capture workflows, validate observations, structure repositories, and surface knowledge to users. Their scope is a unit — a battalion, a brigade, a staff section. When they build well, their unit retains lessons and avoids repeating known mistakes. When they build poorly, the knowledge system goes unused and the unit loses institutional memory at every rotation.

A TM-50E Advanced KM designs the infrastructure within which TM-40E KMs operate. Their scope is a theater — V Corps, 21st TSC, USAREUR-AF headquarters, and the NATO partners they interface with. When they design well, knowledge flows across echelons without requiring manual intervention, lessons from one formation inform the planning decisions of another, and the organization compounds its learning over time. When they design poorly, the result is not a failed system — it is a theater of failed systems, each isolated, each redundant, collectively representing a massive organizational investment in knowledge that cannot be found, shared, or used.

The conceptual shift required is not just one of scale. It is one of purpose.

1-2. From Capture to Learning Infrastructure

Tactical KM — the TM-40E level — is primarily concerned with capture: getting observations into a system before the knowledge walks out the door at RIP/TOA. This is necessary, but it is not learning. A library full of unread books is not an educated organization.

Strategic KM — the TM-50E level — is concerned with the full learning cycle:

Phase	Tactical KM focus	Strategic KM focus
Capture	Build submission workflows	Design taxonomy that makes submissions findable at scale
Validation	Review individual lessons	Design the quality tier architecture and the human review pipeline
Distribution	Surface lessons to unit users	Design cross-domain linkages so logistics lessons reach planners
Application	Notify users of relevant lessons	Instrument the system to detect whether recommendations are used
Doctrine	Escalate high-value lessons	Design the pathway from validated lesson to formal guidance
Retirement	Archive superseded material	Design governance cycles that systematically retire outdated knowledge

The critical insight is that the hardest phase is not capture — it is application. An organization that captures ten thousand lessons and applies none of them has not learned. It has archived. The Advanced KM designs for application as the primary outcome, with capture as a necessary input.

1-3. The Compounding Learning Organization

The goal of enterprise KM is not a repository. It is an organization that gets systematically smarter over time. This is not a metaphor — it is a design specification.

A compounding learning organization has three properties:

Lessons propagate. A lesson learned in V Corps during Saber Strike reaches a planner at 21st TSC before the next exercise. This requires cross-domain linkage, a distribution mechanism, and a taxonomy that makes the lesson findable by someone who did not submit it.

Doctrine incorporates experience. Validated lessons from operations feed into updated SOPs, TTPs, and training guidance. This requires a formal pathway from the KM system to the publication authority, and a publication authority that actually consumes KM outputs.

Rotation cycles do not reset the system. An organization that has to rebuild its knowledge system every 18 months because the KM who built it PCS'd has not compounded anything — it has run in place. Compounding requires designed durability: documentation, succession planning, system architecture that does not depend on a single person's expertise to operate.

The Advanced KM who does not design explicitly for all three of these properties will produce a system that works well for 18 months and fails at the next rotation cycle. That is a predictable outcome, and it is a design failure, not an operational one.

1-4. Vignette: The Defender Europe Lessons Learned Problem

During Defender Europe 22, V Corps collected over 2,400 observations across participating formations. The V Corps KM team validated and classified 847 lessons. Of those, 312 were categorized as high-value — warranting review by adjacent functional areas or higher echelons.

At the conclusion of the exercise, 309 of the 312 high-value lessons remained within the V Corps knowledge system. Three were formally escalated to USAREUR-AF doctrine staff. There was no distribution mechanism to route logistics lessons to 21st TSC, no linkage between operational planning lessons and the JOPP training program at Grafenwoehr, and no coalition-releasable version of any lesson produced.

Defender Europe 23 began planning 14 months later. Planners had no structured access to the Defender 22 lessons. The information existed — but it was not findable, not linked to current planning questions, and not available to coalition partners whose forces would participate in the exercise.

This is not a data problem. It is a KM architecture problem. The Advanced KM's role is to design the system so that Defender Europe 23 planners can, within minutes, surface every relevant logistics, communications, and fires lesson from Defender 22 — regardless of which staff section originally captured it, regardless of security level (with appropriate filtering), and regardless of whether the original submitter is still assigned to USAREUR-AF.

SECTION 2 — ENTERPRISE KNOWLEDGE ARCHITECTURE

2-1. The Scale Problem

BLUF: A knowledge architecture that works for a battalion breaks at Corps. A knowledge architecture that works for a single functional domain breaks when you add cross-domain requirements. Enterprise KM architecture must be designed for scale from the beginning — retrofitting is expensive and frequently fails.

The USAREUR-AF theater presents a specific enterprise KM challenge: multiple echelons (Army, Corps, Division, Brigade), multiple functional domains (G2, G3, G4, G6, G9, Aviation, Special Operations), multiple classification levels, and coalition partner integration requirements — all of which must be served by knowledge systems that are consistent enough to share information but flexible enough to meet the distinct requirements of each organization.

The naive solution is a single centralized repository. The theater KM builds one system, everything flows in, users search and find. This fails for a predictable set of reasons: no single taxonomy satisfies all functional domains, centralized review becomes a bottleneck at scale, units stop submitting when their

specific requirements are not met, and the system becomes dominated by whatever functional area has the most aggressive KM.

The other naive solution is full decentralization: every unit builds its own system, uses its own taxonomy, and owns its own repository. This produces knowledge silos. Cross-domain and cross-echelon knowledge sharing requires manual coordination between KM custodians, which means it rarely happens.

The correct approach is federated architecture.

2-2. Federated vs. Centralized: The Design Choice

Characteristic	Centralized	Federated	Recommended for USAREUR-AF
Taxonomy control	Single authority	Distributed with shared core	Federated with mandatory core
Review/validation	Central KM team	Unit KMs with central coordination	Federated
Cross-domain discovery	Easy (same system)	Requires linkage design	Design linkages explicitly
Unit autonomy	Low	High	High, with governance guardrails
Coalition integration	Single interface point	Multiple interface points	Centralized interface, federated storage
Resilience to personnel loss	Low (single point of failure)	High	Federated
Consistency of quality	High (if central team is strong)	Variable	Requires quality tier standards

The USAREUR-AF architecture should be federated at the storage and curation level, with a mandatory shared taxonomy core that enables cross-domain and cross-echelon search. Each major formation (V Corps, 21st TSC, USAREUR-AF HQ) maintains its own knowledge repository on MSS. Each repository uses the shared taxonomy core for classification. A Theater Knowledge Index aggregates metadata — not content — from all repositories, enabling theater-wide search without centralizing content or review authority.

2-3. Designing the Shared Taxonomy Core

The shared taxonomy core is the minimum set of classification dimensions that every knowledge product in the theater must carry. It enables cross-domain search without requiring every unit to adopt the same internal taxonomy.

For USAREUR-AF, the shared core should include:

Dimension	Required values	Notes
Functional domain	G1, G2, G3, G4, G6, G9, Aviation, Fires, Sustainment, Medical, Legal, CEMA, SOF, Multi-domain	Units may add sub-domains; must map to at least one core domain
Echelon	Theater, Corps, Division, Brigade, Battalion	Echelon at which the lesson is applicable
Exercise / operation	Defender Europe, Saber Strike, Combined Resolve, Swift Response, [named operation], Steady-state	Source event or context
Quality tier	Observation, Reviewed, Validated, Doctrine	See Section 4
Releasability	US-only, NATO-releasable, PfP-releasable, Public	See Section 6
Status	Active, Under review, Superseded, Retired	See Section 8

Units may extend their internal taxonomies as needed. What they may not do is omit core dimensions — the Theater Knowledge Index depends on consistent core classification to enable cross-unit search.

2-4. The Theater Knowledge Index on MSS

The Theater Knowledge Index is not a repository. It is a metadata aggregator — an MSS dataset that contains the core classification fields, a title, a summary, a URL back to the source repository, and the access control level for each knowledge product across all federated repositories.

Users search the Theater Knowledge Index to discover relevant knowledge across the theater. When they find a relevant item, the index points them back to the originating unit's repository for full content access (subject to access control). The KM who manages the Theater Knowledge Index is responsible for taxonomy consistency and index freshness — not for content ownership.

This architecture has a critical property: no single unit's knowledge is replicated into a central system it does not control. Each unit remains the authoritative owner of its own knowledge products. The theater-level system provides discovery without requiring custody transfer.

2-5. Vignette: V Corps and 21st TSC — One Theater, Two Architectures

V Corps operates a knowledge management system centered on operational lessons: fires synchronization, ISR allocation, command post operations, information operations. Their taxonomy is G-staff focused, exercise-driven, and deeply integrated with the JOPP review cycle.

21st TSC operates a knowledge management system centered on sustainment lessons: distribution network design, maintenance backlogs, medical logistics, contractor support integration. Their taxonomy is functional-area focused, tied to logistics readiness cycles, and integrated with LOGSTAT reporting.

Both formations are in the USAREUR-AF theater. A lesson about fuel distribution during Defender Europe 22 — captured in 21st TSC's system under "POL distribution, forward" — is directly relevant to V Corps planners designing fuel support for Defender Europe 23. But without a shared taxonomy that maps 21st TSC's "POL distribution" tag to the theater-level "Sustainment" domain, and without a Theater Knowledge Index that makes 21st TSC lessons visible to V Corps search queries, that lesson never reaches the people who need it.

The shared taxonomy core and the Theater Knowledge Index together solve this problem without requiring either formation to restructure its internal KM system.

SECTION 3 — AI-ASSISTED KNOWLEDGE MANAGEMENT — THE MSS OPPORTUNITY

3-1. The Role of AI in KM Workflows

BLUF: AIP Logic and AI tools on MSS can substantially reduce the manual workload of knowledge management at scale. They can also introduce systematic quality failures if deployed without appropriate human oversight. The design principle is invariant: AI augments the KM workflow; it does not replace the human judgment required to validate and approve knowledge products.

At theater scale, the KM workflow generates volumes that human reviewers cannot process at the rate of submission. A Corps-level exercise may generate 200 to 400 observations in 72 hours. A theater-level exercise program running across multiple concurrent exercises may generate that volume daily. Manual review at that rate is not feasible without either a large dedicated KM staff or significant review latency — both of which create operational problems.

AI augmentation addresses this by handling the high-volume, low-judgment portions of the workflow: classification, deduplication, similarity detection, and initial routing. Human reviewers are then focused on the high-judgment portions: validation, approval, and doctrine escalation decisions.

3-2. AI Augmentation Functions in KM Workflows

Function	What AI does	Human role	Risk if human role is removed
Auto-tagging	Suggests taxonomy classifications based on submission content	Review and confirm; correct systematic errors	Systematic misclassification; broken cross-domain discovery
Duplicate detection	Flags submissions that appear similar to existing knowledge products	Review flagged pairs; decide whether to merge, link, or keep separate	Duplicate accumulation or inappropriate merging of distinct lessons
Pattern detection	Identifies recurring themes across large observation sets	Interpret patterns; decide whether pattern represents a systemic issue warranting escalation	False patterns; real patterns missed because AI framing is accepted without interrogation
Precedent recommendation	Surfaces related validated lessons when a new submission arrives	Review recommendations; decide whether to add cross-references	Irrelevant links that reduce system signal-to-noise; relevant links missed
Summary generation	Produces plain-language summaries of complex technical lessons	Review and edit; ensure accuracy and appropriate classification	Inaccurate summaries distributed as authoritative; classification errors
Draft SOP/TTP generation	Produces structured draft guidance from validated lesson sets	Full review, edit, and approval by qualified SME and appropriate authority	AI-generated doctrine published as authoritative; errors propagate into training and operations

The pattern is consistent: AI handles first-pass processing; humans handle all approval decisions. The speed benefit comes from reducing the cognitive load on human reviewers — they are adjudicating AI proposals rather than building from scratch, which is substantially faster without sacrificing judgment.

3-3. Designing the AI-Augmented Review Queue

The practical implementation of AI augmentation in an MSS KM system is the AI-augmented review queue. Submitted observations enter the queue. An AIP Logic pipeline processes each submission and adds:

- Suggested taxonomy tags with confidence scores
- Duplicate detection flags (if applicable)
- Linked related lessons (top 3-5 by relevance score)

- A draft one-sentence summary

The human reviewer sees the submission plus the AI-generated metadata. They confirm or modify taxonomy, resolve duplicate flags, accept or reject related-lesson links, and edit the summary. Review time for a typical observation drops from 15-20 minutes (manual) to 3-5 minutes (AI-augmented review). At Corps scale, this is the difference between a team of 6 KMs managing the review queue and a team of 2.

NOTE

The confidence score on AI-suggested tags is a useful signal, not a threshold for auto-approval. Do not design workflows where high-confidence AI tags are automatically accepted without human review. The cost of systematic misclassification at theater scale — thousands of improperly tagged lessons — exceeds the labor cost of review.

3-4. Pattern Detection as an Intelligence Function

One of the highest-value AI augmentation functions is pattern detection across large observation sets. This is qualitatively different from the other functions because it does not operate on individual submissions — it operates on the corpus.

An AI pipeline that analyzes 2,000 submitted observations from a year of exercises and identifies "fuel distribution failures at forward support companies" as a recurring theme in 14% of logistics submissions is surfacing something that no individual reviewer would detect. This is genuine organizational intelligence — the kind of insight that, acted on, improves planning and prevents repeated failures.

The Advanced KM designs for this function deliberately. This means maintaining corpus quality (garbage in, garbage out), designing queries that surface recurring themes rather than just individual keywords, and establishing a governance process for acting on identified patterns — who reviews the pattern report, who decides whether to escalate to doctrine, and who tracks whether the recommended changes were actually made.

Pattern detection results should never be auto-published. A recurring theme in AI analysis may reflect a genuine systemic problem, a systematic bias in who submits observations, a classification error pattern, or an artifact of how the AI is processing language. Human analysis is required to distinguish among these before any pattern-based recommendation is issued.

DRAFT

SECTION 4 — KNOWLEDGE QUALITY ARCHITECTURE

4-1. The Quality Problem at Scale

BLUF: Not all submitted knowledge is equal. At unit scale, a KM can maintain quality through direct involvement in every submission. At theater scale, direct involvement is impossible — quality must be designed into the system architecture.

A theater-scale knowledge system that treats a rough field observation with the same authority as a doctrine-reviewed SOP is a system that will be ignored. Users develop trust in knowledge systems through experience — if early searches surface authoritative, useful results, users return. If early searches surface unvalidated, contradictory, or outdated material, users stop searching and the system becomes an unused archive.

Quality architecture solves this by making quality visible and searchable. Users can filter by quality tier, calibrate their trust accordingly, and know that Validated lessons have been reviewed by a qualified SME with appropriate authority.

4-2. The Quality Tier Model

Tier	Label	Definition	Required approver	Search priority
Tier 0	Observation	Raw submission; not yet reviewed	None (submitter only)	Lowest; filtered out by default for general search
Tier 1	Reviewed	Checked for completeness, accuracy, and relevance by a subject matter expert	Unit-level SME	Available in search; displayed with tier label
Tier 2	Validated	Approved by a responsible authority (section chief, S3/G3 branch, equivalent)	O-5 or above, or designated program authority	Prioritized in search results
Tier 3	Doctrine	Incorporated into formal guidance (SOP, TTP, EXORD annex, published doctrine)	Publication authority (varies by document type)	Highest priority; distinguished by tier in all search results

The quality tier is a field in the core taxonomy and a search filter in every Workshop application. The default search view for operational users surfaces Tier 2 and above. KMs and SMEs can access Tier 0 and 1 for review work. All tiers are available to Advanced KMs managing the pipeline.

DRAFT

4-3. The Bottleneck Problem

Quality review is a bottleneck by design — human judgment cannot scale linearly with submission volume. The Advanced KM's job is not to eliminate the bottleneck (that would eliminate quality) but to design for throughput without sacrificing integrity.

Five design levers control throughput:

Distributed review authority. Do not centralize all validation in a single KM or a single office. Design the review workflow so that functional domain SMEs handle validation within their domains. A sustainment SME validates logistics lessons; a fires SME validates fires lessons. The KM coordinates the pipeline; they do not validate all content personally.

Tiered review thresholds. Not every observation requires full validation to be useful. A Tier 1 reviewed observation — checked for completeness and accuracy by a competent SME — is useful for a planner who understands they are working with reviewed-but-not-validated material. Reserve the full validation process (Tier 2) for lessons that are likely to be widely shared, cited in plans, or escalated to doctrine.

AI pre-processing to reduce reviewer load. As described in Section 3, AI augmentation of the review queue reduces per-submission review time substantially. This multiplies the throughput of a fixed reviewer population without reducing review rigor.

Review cycle batching. For non-time-sensitive submissions, batch reviews by functional domain and schedule regular review cycles (weekly or bi-weekly). This reduces context-switching cost for SME reviewers who are performing KM review duties as a collateral function.

Queue visibility. A review queue that is not visible is a queue that stalls. Build dashboards that show review queue depth, average review time by domain, and submissions aging past threshold. This makes the bottleneck visible to program leadership, which is the prerequisite for resourcing decisions to address it.

4-4. Integrity Without Bottleneck: The Advanced KM's Balance

The tension between throughput and integrity is never fully resolved — it is managed. The Advanced KM's role is to calibrate the system to the operational tempo: tighter review gates during steady-state operations, faster throughput with appropriate labeling during high-tempo exercises. The quality tier model supports this flexibility because it communicates confidence level to users without hiding the material. A planner working under time pressure can choose to use a Tier 1 reviewed observation with appropriate caution. A doctrine writer developing a new SOP should work exclusively from Tier 2 and Tier 3 material.

The integrity risk is not the planner making an informed choice to use Tier 1 material. The risk is a system where quality tiers are inconsistently applied, where users cannot trust that a Tier 2 label means what it says, or where AI pre-processing has effectively replaced human validation without the design explicitly acknowledging that fact. Those are failure modes, not features.

SECTION 5 — CROSS-DOMAIN KNOWLEDGE INTEGRATION

5-1. The Silo Problem

BLUF: Knowledge silos are the default outcome of any enterprise KM system that does not explicitly design for cross-domain integration. Every staff section with a SharePoint folder, every unit with its own Foundry dataset, every functional area with its own taxonomy is, in isolation, building a knowledge silo. The Advanced KM designs the linkages that make cross-domain knowledge discoverable.

The operational cost of knowledge silos is difficult to quantify but easy to recognize in retrospect. A sustainment lesson about the failure mode of a particular fuel distribution configuration during cold-weather operations is captured in the 21st TSC KM system. An operational planner at V Corps designs a cold-weather operation with a fuel support configuration that replicates the same failure mode — because they had no way to know the lesson existed. The lesson was captured. The learning did not happen.

Cross-domain integration does not require merging repositories, adopting a unified taxonomy, or centralizing review authority. It requires three design elements: shared classification dimensions, linked Object Types in the Ontology, and cross-domain search access.

5-2. Designing for Cross-Domain Discoverability

The shared taxonomy core (described in Section 2) provides the first design element. When every knowledge product carries a functional domain tag from a shared controlled vocabulary, a search for "Sustainment" lessons will surface relevant knowledge products from every repository in the theater that uses the shared core — regardless of each unit's internal taxonomy.

The Ontology provides the second design element. In MSS, Object Types can be linked: a Lesson Object Type can have a "related lessons" link to other Lesson objects, and a "source exercise" link to an Exercise object. When a G4 planner views a sustainment lesson, a properly linked Ontology shows related operational planning lessons from the G3 domain — because a KM or an AI pipeline has explicitly created those relationships. The Ontology makes the cross-domain connection navigable.

The third element is access. Cross-domain search is only useful if the user performing the search has read access to the relevant repositories. The Advanced KM designs access control to be permissive for search (broad read access to metadata and summaries) and appropriately controlled for full content (access tiered by classification and need-to-know). The Theater Knowledge Index model from Section 2 implements this: index metadata is broadly accessible, full content access is gated at the originating repository level.

5-3. The Cross-Domain Link as a KM Work Product

Explicit cross-domain links do not create themselves. Someone must decide that a sustainment lesson about fuel distribution failure modes is relevant to an operational planning lesson about tempo sustainability and create the link between them. This is KM work, and it requires both functional domain knowledge and an understanding of the Ontology structure.

At theater scale, the Advanced KM cannot personally create every cross-domain link. The design solution is distributed cross-linking: when a KM validates a lesson in their domain, they are responsible for adding cross-domain links to related lessons in other domains. AI-suggested related lessons (from the augmented review queue, Section 3) reduce the discovery work — the KM reviews suggestions and confirms or rejects them. The result is a progressively richer cross-domain link structure that reflects the judgment of distributed functional SMEs rather than the workload capacity of a single theater KM.

5-4. Vignette: The Fires-Sustainment Link That Prevented a Planning Error

During planning for a V Corps combined arms live fire, the G3 fires planner queried the theater knowledge system for relevant lessons on artillery ammunition resupply during high-tempo fires operations. The system returned eight results — six from Fires domain sources and two from Sustainment domain sources.

The two Sustainment domain results — cross-linked by a 21st TSC KM who had identified the relevance during validation — contained a Tier 2 validated lesson on ammunition convoy throughput constraints during contested road network conditions. The lesson was specific about the throughput numbers that the 21st TSC had experienced during Defender Europe 22: 40% lower than planning factors assumed.

The fires planner used those throughput constraints to adjust the planned fires tempo. The revised plan required a larger pre-positioned ammunition stockpile to compensate. The sustainment requirement was identified during planning — not during execution.

Without the cross-domain link, the planner would have had no way to know the 21st TSC lesson existed. The lesson was in a different repository, tagged under a different functional domain, and would not have appeared in a fires-only search. The cross-domain link was the single design element that converted a captured lesson into a planning decision.

DRAFT

SECTION 6 — COALITION KNOWLEDGE MANAGEMENT

6-1. The Coalition KM Challenge

BLUF: USAREUR-AF does not operate alone. Coalition partners — NATO member nations, Partnership for Peace nations, and bilateral partners — contribute forces to exercises and operations throughout the theater. Coalition KM requires designing a system that serves both US-only and coalition-releasable knowledge without creating two separate systems.

The default response to coalition knowledge sharing is to create a separate coalition system — a parallel repository with a separate taxonomy, separate review workflow, and separate user population. This is administratively clean and operationally counterproductive. It doubles the KM workload, creates divergent knowledge bases that are difficult to reconcile, and produces coalition partners who have access to a curated fraction of theater knowledge rather than a genuinely useful operational resource.

The alternative is a releasability-tiered single system: one knowledge repository with a releasability tag on every knowledge product that controls who can access it. US-only knowledge is not visible to coalition partners. Coalition-releasable knowledge is accessible to partner nation users with appropriate credentials. The KM creates one knowledge product; the system handles access control.

6-2. The Releasability Tag as a Design Element

The releasability tag is part of the mandatory shared taxonomy core (Section 2) and is enforced at the Foundry dataset and Workshop application level through MSS access controls. The four-level model used across USAREUR-AF:

Releasability level	Accessible to	Default for new submissions	Modification authority
US-only	US personnel with appropriate clearance	Yes — default for all new submissions	Originating unit KM, with OPSEC review for downgrade
NATO-releasable	All NATO member nation forces	No — requires explicit designation	Originating unit KM, with OPSEC and classification review
PfP-releasable	Partnership for Peace nation forces	No — requires explicit designation	Originating unit KM, with OPSEC, classification, and legal review
Public	Unrestricted	No — requires explicit designation and public affairs review	Designated publication authority only

WARNING

Downgrading releasability — particularly to NATO-releasable or PfP-releasable — is not a routine KM administrative action. It requires OPSEC review to ensure that the aggregated knowledge product does not reveal sensitive capability data, operational patterns, or personnel information even if individual elements are individually unclassified. A coalition partner does not need to know that USAREUR-AF's medical logistics system has a specific throughput bottleneck to understand a general lesson about medical logistics planning. The Advanced KM must design for content sanitization as part of the coalition-releasable designation workflow.

6-3. STANAG 4778 and NATO Information Management Policy

Coalition knowledge sharing operates under a governance framework that the Advanced KM must understand and design within. STANAG 4778 governs NATO lessons learned interoperability. NATO Information Management Policy governs what information can be shared across the Alliance and through what channels.

For practical KM system design purposes, the key constraints are:

Information sharing agreements must precede technical access. A partner nation cannot be given access to coalition-releasable content on MSS without a signed information sharing agreement on file with the appropriate authority. The KM does not manage information sharing agreements — but the KM must not grant access without confirming that the agreement exists.

Coalition knowledge products must be explicitly designated. No knowledge product is coalition-releasable by default. The US-only default ensures that knowledge products require a deliberate decision to release — protecting against inadvertent disclosure through system misconfiguration.

The architecture must support auditability. Coalition sharing requires the ability to demonstrate, after the fact, what was shared with whom and when. Foundry's access control and audit logging capabilities support this, but the KM must design the system to preserve those logs and ensure they are accessible for review.

6-4. Designing for Coalition Without Creating Two Systems

The single-system, releasability-tiered architecture requires specific design decisions in MSS:

Separate Workshop views by releasability. US personnel see the full system. Coalition partner users see a Workshop application filtered to their releasability level. The underlying data is the same; the view is access-controlled. This requires designing the Workshop applications from the beginning to support releasability filtering as a core parameter.

Coalition-appropriate metadata. A knowledge product designated as NATO-releasable may require a sanitized title and summary that do not reference US-only context. The KM workflow should include a step to review and edit metadata for coalition-appropriate language when designating a product as coalition-releasable.

Coalition feedback integration. If coalition partners are submitting observations into the system — which is valuable and should be encouraged — their submissions carry a source nation tag and their own releasability designation. A lesson submitted by a German Bundeswehr KM is not automatically releasable to all NATO partners; it carries whatever releasability designation the submitting nation assigns. The theater KM system must support multi-nation releasability tags, not just a US-centric model.

SECTION 7 — MEASURING KNOWLEDGE SYSTEM EFFECTIVENESS

7-1. Beyond Count Metrics

BLUF: Submission volume, repository size, and taxonomy completeness are process metrics. They measure KM system activity, not KM system effectiveness. An Advanced KM who reports "2,400 observations submitted" without also reporting whether those observations were found and used by the people who needed them has measured the input, not the outcome.

Process metrics are not worthless. They are necessary for managing KM system operations and identifying bottlenecks. But they are insufficient for demonstrating that the knowledge system is producing operational value. The Advanced KM needs a measurement architecture that connects KM system activity to operational outcomes.

DRAFT

7-2. The Effectiveness Metric Framework

Metric type	What it measures	How to instrument on MSS	Operational interpretation
Submission volume	Observation capture rate	Count of Observation objects by time period, domain, unit	Baseline activity; high volume does not indicate effectiveness
Review throughput	Quality tier advancement rate	Count of tier transitions (Obs → Reviewed → Validated) by time period	KM pipeline health; persistent backlogs indicate resource constraint
Search utilization	Whether users are querying the system	Workshop analytics: search query counts, unique users by week	Adoption rate; zero search means zero utility regardless of content quality
Search result engagement	Whether users are opening results	Workshop analytics: result click-through rate by tier and domain	Content quality signal; low engagement after search suggests results are not relevant
Cross-domain discovery	Whether cross-domain links are being followed	Workshop analytics: navigation events from one domain to another	Cross-domain integration effectiveness
Lesson citation	Whether lessons are cited in plans or products	Reference tracking: links from plan documents or other knowledge products to validated lessons	The highest-value outcome metric; requires a citing convention and compliance
Repeated failure detection	Whether the organization is making the same mistakes twice	Periodic analysis: do new observations match themes of validated lessons in the same domain	The definitive organizational learning metric; requires periodic human analysis of patterns
Rotation continuity	Whether system usage continues through RIP/TOA	Search volume and submission rates during and after unit transitions	System durability metric; drops at transition indicate person-dependent systems

7-3. Foundry Workshop Analytics

MSS Foundry Workshop includes native usage analytics that provide search query volumes, user engagement, and navigation patterns. The Advanced KM should configure a KM Effectiveness Dashboard in Workshop that surfaces the key metrics from the table above on a weekly or monthly review cycle.

DRAFT

The analytics data is most useful when interpreted in context. A drop in search utilization the week after a major exercise ends is expected — the operational tempo is lower. The same drop three months into steady-state operations, with no corresponding decrease in submission volume, suggests that users submitted but did not engage with the system to retrieve. That is the pattern that warrants investigation: the supply side of the KM system is functioning, but the demand side is not.

7-4. The Repeated Failure Metric

The most operationally significant metric in the framework — whether the organization is making the same mistakes twice — is also the most difficult to instrument because it requires human interpretation. No automated pipeline can reliably determine whether a new observation represents a repeated failure or a similar-but-distinct situation.

The practical implementation is a quarterly analysis cycle: the Advanced KM (or designated theater KM analyst) reviews newly validated lessons against the validated lessons corpus, identifies thematic matches, and produces a brief report: "These five new validated lessons appear to represent recurrences of previously validated lessons in the same domain. Recommended action: review the prior lessons to determine if they were incorporated into planning, training, or doctrine, and if not, escalate for action."

This analysis does not require AI. It benefits from AI-assisted similarity detection (flagging candidate matches for human review) but the judgment — is this the same failure? — is a human judgment. The value of the metric depends on the quality of that judgment, which is why it belongs at the Advanced KM level.

SECTION 8 — KNOWLEDGE LIFECYCLE MANAGEMENT AT SCALE

8-1. The Accumulation Problem

BLUF: A knowledge system that never retires outdated material will eventually become an obstacle to knowledge access. Users who cannot distinguish current lessons from superseded ones, current SOPs from outdated ones, and active lessons from historical archives will stop trusting search results — and the system will become a liability rather than an asset.

At unit scale, the KM can manage currency through direct familiarity with the repository. At theater scale, no individual KM can maintain that familiarity. Currency management must be designed into the governance architecture.

The accumulation problem compounds over time. A theater KM system that has been operating for five years without systematic retirement processes contains knowledge products from organizational structures that no longer exist, equipment that has been replaced, techniques that have been superseded

by updated doctrine, and observations from exercises that have been thoroughly analyzed and incorporated into current guidance. That material is not wrong — it may be historically interesting — but its presence in active search results reduces the signal-to-noise ratio for users seeking current applicable guidance.

8-2. Knowledge Lifecycle Stages

Stage	Status tag	Description	Trigger	KM action
Active	Active	Current, applicable knowledge	Default on validation	None — maintain
Under review	Under review	Flagged for currency or accuracy review	Scheduled review cycle, or flagged by user or AI	Assign to reviewing SME
Superseded	Superseded	Replaced by a newer knowledge product or updated doctrine	Reviewer determination	Link to superseding product; remove from default search
Retired	Retired	No longer applicable; archived for historical reference	Reviewer determination after supersedure period	Move to archive tier; accessible to KMs, not in operational search

The lifecycle stage is a status field in the shared taxonomy core and controls default search visibility: Active and Under review products appear in operational search; Superseded and Retired products do not (but remain accessible to users who explicitly filter for them or who navigate to the archive).

8-3. The Scheduled Review Cycle

The governance mechanism that drives lifecycle management is the scheduled review cycle: a calendar-driven process that systematically reviews knowledge products whose age or triggering conditions indicate potential currency issues.

A baseline review schedule for a theater KM system:

DRAFT

Review type	Frequency	Scope	Responsible party
New validation review	Continuous	Tier 1 observations ready for Tier 2 validation	Domain SMEs per KM assignment
Annual currency review	Annual	All Tier 2 validated lessons older than 24 months	Theater KM, with domain SME review
Post-doctrine update review	Event-driven	All lessons in domains affected by a doctrine publication change	Theater KM; triggered by doctrine publication events
Post-equipment change review	Event-driven	All lessons referencing equipment undergoing fielding or replacement	Theater KM; triggered by G4 equipment transition notifications
Post-RIP/TOA review	Event-driven	All lessons associated with the transitioning unit	Outgoing unit KM, with theater KM oversight
Quarterly pattern review	Quarterly	New validated lessons vs. existing corpus for recurrence analysis	Theater KM or designated analyst

The post-RIP/TOA review is the highest-risk event for knowledge currency. When a unit transitions, the organizational context that gave meaning to many of their knowledge products transitions with them. Unit-specific references, named individuals, internal system names, and exercise-specific context may no longer be interpretable by the incoming unit. The KM lifecycle review at RIP/TOA should address whether unit-specific knowledge products are still meaningful in the incoming unit's context, or whether they should be generalized, superseded, or retired.

8-4. Designing for Three Rotation Cycles

A durable KM system is one that still functions well after three RIP/TOA cycles — approximately 4-5 years — without significant architectural intervention. Designing for this durability requires explicit attention to person-dependence at every design decision.

A person-dependent KM system is one where: - System operation requires knowledge that lives primarily in the current KM's head rather than in documented procedures - Taxonomy decisions were made informally and exist as institutional knowledge rather than written governance - Cross-domain links were created by one person and are not understood by successors - AI pipeline configurations are undocumented and require the original architect to maintain - Review authority relationships are personal rather than structural (e.g., "ask MAJ Smith" rather than "submit to G3 KM branch chief")

None of these properties are inevitable. They result from design choices made under time pressure, or from KMs who did not document their work. The Advanced KM designs explicitly against them:

- Every governance decision is documented as a written policy or architecture decision record (ADR)
- Every taxonomy element has a written definition in the shared governance documentation

- Every AI pipeline has a runbook: what it does, how it is configured, how to modify it, who to contact when it fails
- Every review authority relationship is tied to a position, not a person
- Every system has a "keys in hand" handover package: current as of the last rotation, updated as part of every RIP/TOA process

The test for three-rotation durability is simple: if the current Advanced KM were reassigned tomorrow, could their replacement understand and maintain the system from documentation alone? If the answer is no, the system is not yet durable.

SECTION 9 — ADVANCED FAILURE MODES — WHAT TM-50E KMS GET WRONG

9-1. Why Advanced KMs Fail Differently

BLUF: Advanced KM failures are not the same as TM-40E failures. A TM-40E KM who builds a poor system affects a unit. A TM-50E KM who builds a poor system affects a theater. The failures are also more subtle — sophisticated architecture can obscure fundamental problems until the system is deployed at scale, at which point remediation is expensive.

The following failure modes are specific to the Advanced KM level. They are not hypothetical — each represents a pattern observed in enterprise knowledge management programs across large organizations.

9-2. The Architecturally Impressive, Operationally Unused System

The most common advanced failure: a KM system that is technically sophisticated, beautifully designed, and virtually unused by operational personnel. The failure mechanism is optimization for the KM's priorities (completeness, taxonomy consistency, architectural elegance) at the expense of the user's priorities (speed, relevance, simplicity).

Indicators: - Submission forms with more than 15 fields - Search interfaces that require knowledge of the taxonomy to use effectively - Knowledge products that are accurate but written for a KM audience, not an operational audience - System adoption metrics that were never built because the design team assumed adoption would follow quality

Correction: Design for the end user first. The KM's job is to make operational personnel more effective. If the people who need the system are not using it, the system has failed — regardless of its technical quality.

9-3. Designing for Capture Without Designing for Retrieval

A system that is easy to submit to but hard to search is a well-organized archive. It accumulates knowledge that cannot be found when needed. The failure is prioritizing the supply-side workflow (submission, validation, storage) over the demand-side workflow (search, discovery, application).

Indicators: - More design effort invested in the submission form than in the search interface - No analytics on search utilization or result engagement - Users who know lessons exist but cannot find them - Search results that return accurate results but in relevance order that surfaces outdated material ahead of current material

Correction: Spend at least as much design effort on retrieval as on capture. Test the search experience with operational users before deployment. Instrument for retrieval effectiveness from day one.

9-4. AI Augmentation Creating False Confidence in Knowledge Quality

AI-assisted KM workflows reduce review workload — and if not carefully designed, they reduce human scrutiny along with it. When reviewers become accustomed to confirming AI-suggested tags rather than independently evaluating submissions, the human reviewer is no longer functioning as an independent quality check. They are auditing the AI's work, and if the AI has a systematic bias or error mode, the reviewer is likely to miss it.

Indicators: - AI tag confirmation rates above 95% (near-perfect agreement may indicate reviewers are not independently evaluating) - Systematic misclassification patterns that persist across the validation pipeline - Reviewers who cannot articulate why a submission was classified the way it was - AI-generated summaries published without material human editing

Correction: Design the review workflow to require independent human evaluation before displaying AI suggestions, or at minimum to include a mandatory "disagree" step that requires active confirmation rather than passive acceptance. Regularly audit AI suggestion accuracy against independent human classification of the same submissions. Train reviewers to interrogate AI proposals, not ratify them.

9-5. Failing to Sustain Organizational Commitment Through Rotation Cycles

A KM system requires sustained organizational commitment — leadership attention, resourcing, and active use by operational personnel — to remain effective. When the champion of the system PCS's, when leadership priorities shift, or when a high-operational-tempo period draws personnel away from KM functions, usage drops. If the system is not resilient to these periods, recovery is difficult.

The failure mode is building a system whose health depends on the sustained attention of a specific individual or a specific leadership priority. When that individual or priority changes, the system degrades, submissions fall behind, review backlogs accumulate, and operational users who encounter stale or sparse results stop searching — accelerating the degradation.

DRAFT

Correction: Design for minimum viable operation. Identify the absolute minimum KM resource level at which the system can maintain acceptable quality without degradation, document it as a resourcing requirement, and brief it to leadership as a sustainment requirement (not a capability request). A system that needs one skilled KM and four hours per week to maintain its minimum viable state is a system that leadership can commit to. A system that needs continuous attention from a team of specialists is a system that will fail at the first operational tempo spike.

9-6. Building Knowledge Systems Without Decommission Plans

Every knowledge system will eventually be superseded — by a new platform, a new architecture, a new organizational requirement, or a new classification of information. A system without a documented decommission plan will outlive its operational utility and become a liability: users uncertain whether the old system or the new system is authoritative, migration costs that exceed original build costs, and governance conflicts between custodians of the old and new systems.

The decommission plan is not a prediction — it is a design artifact that forces the Advanced KM to articulate: when would this system be appropriately retired? What are the conditions that would trigger a transition? What does data migration look like? Who owns the archive?

Documenting this at design time does not mean the system will be decommissioned soon. It means that when the time comes, the decommission is a managed process rather than an emergency.

SUMMARY — KEY PRINCIPLES FOR THE ADVANCED KM

The following principles synthesize the conceptual framework of this guide. They are not rules — they are the professional orientation of an Advanced KM operating at theater scale.

Design for application, not capture. The measure of a knowledge system is whether its users make better decisions because of it. Capture is a means, not an end.

Federated with a shared core. Theater-scale KM works best when each organization owns its knowledge and a shared taxonomy enables cross-domain discovery. Enforce the core; leave the rest flexible.

AI augments; humans decide. Every AI function in a KM workflow has a corresponding human approval step. The speed benefit comes from reducing cognitive load, not from removing human judgment.

Quality must be designed, not assumed. Build quality tiers into the architecture. Make quality visible to users. Design the review pipeline for throughput, not just rigor.

Cross-domain links are a work product. Discovery across functional domains does not happen automatically. Someone has to create the links. Design the workflow to distribute that work across functional SMEs.

Coalition is one system, not two. The releasability tag enables coalition sharing without duplicating effort. Design for it from the beginning.

Measure outcomes, not activity. Submission counts measure effort. Lesson citation rates, repeated failure detection, and rotation continuity measure effectiveness.

Retire actively. An active retirement program is as important as an active capture program. A repository that never retires material degrades in signal-to-noise ratio until it becomes unusable.

Design for your third successor. Document everything. Tie authority to positions, not people. If the system requires you personally to function, it will fail at your next assignment.

CONCEPTS GUIDE — TM-50E COMPANION | ADVANCED KNOWLEDGE MANAGER | MAVEN SMART SYSTEM (MSS) HEADQUARTERS, UNITED STATES ARMY EUROPE AND AFRICA | Wiesbaden, Germany | 2026 DISTRIBUTION RESTRICTION: Approved for public release; distribution is unlimited.

DRAFT